Homework 1 – Image Enhancement

Computer Vision · Academic Year 2026-27
University of Verona · Master's Degree in Computer Engineering for Intelligent Systems
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1. Objective

This report documents an *image enhancement* pipeline built with classical methods only (OpenCV and NumPy) and applied to five low-contrast photographs with sharp shadows or non-uniform illumination. The goal is not to reconstruct the scene exactly, but to compare four fixed-parameter approaches, observe their effects, and assess the trade-off between detail visibility, noise and color naturalness.

To make the comparison reproducible, each original image was artificially degraded (reduced contrast and Gaussian noise). **The original images serve as the reference** for measuring the synthetic degradations; in a real scenario a clean reference is normally not available, so PSNR, SSIM and MAE measure closeness to the *original*, not perceived quality. The restoration is also *blind*: the methods receive only the degraded image and know neither the contraction factor nor the noise intensity.

2. Data and synthetic degradation

The set comprises five color images chosen because they exhibit at least one of the required conditions (low contrast, shadows or non-uniform illumination, barely visible details). The evaluation set does not consist of synthetic images: the latter are only the degraded version of real photographs. Originals, degraded images and results are in the repository.

Table 1 *– Images of the evaluation set.*

File	Subject and condition	Resolution	Source and license
img.png	White wall with sharp shadows (non-uniform illumination)	3931×5897	Unsplash · Unsplash License
img2.png	Orange sunrise over a hazy valley (low contrast, highlights)	4032×3024	Unsplash · Unsplash License
img3A.png	Night path with fog and street lamps (very dark scene)	4000×6000	Unsplash · Unsplash License
img4.png	Yellow wall with sharp shadows and backlit areas	4389×6584	Unsplash · Unsplash License
img5.png	Mountains with fog (intrinsically low contrast)	7728×5152	Unsplash · Unsplash License

For each original f (BGR, uint8) the degraded version g is obtained with a contrast contraction around the mid-gray c plus independent per-channel Gaussian noise, added in float32, rounded and clipped to [0, 255]:

$$g = \mathrm{clip}_{[0,255]}(\,\alpha\,(f - c) + c + n\,), \quad n \sim \mathrm{N}(0, \sigma^2)$$

with $\alpha = 0.5$, $c = 128$, $\sigma = 10$ and random generator seed 42, identical for all images. The degraded images are saved as PNG so as not to introduce a second degradation from JPEG compression.

3. Methods and parameters

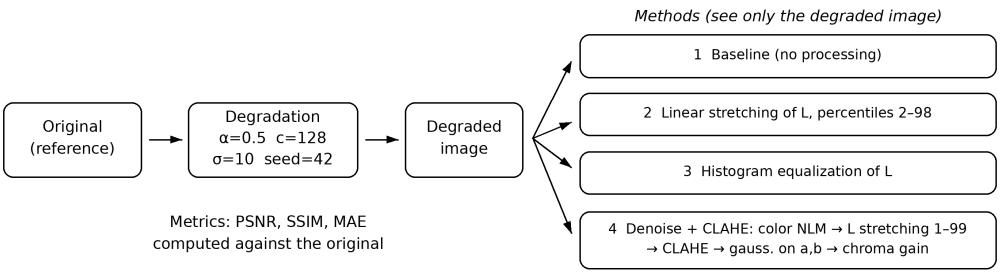


Figure 1 *– Experiment diagram. The methods process only the degraded image; the metrics compare the result with the original.*

1. Baseline. The degraded image with no processing: a method is useful only if it beats it.

2. Linear contrast stretching on luminance. Only the L channel of CIE-Lab is processed, linearly mapping the 2nd and 98th percentiles to 0 and 255; a and b do not change, so the hue remains unchanged. Compared with per-channel RGB stretching (which alters the B:G:R ratios and can cause color casts), color casts are avoided, but values saturated by percentile clipping lose detail.

3. Histogram equalization on luminance. `cv2.equalizeHist` on L (CIE-Lab), with a and b unchanged. The transformation is nonlinear and distributes the levels uniformly: it therefore tends to amplify noise in flat areas and to shift the mean brightness considerably.

4. Additional method: Denoise + CLAHE (motivated combination). Low contrast should be corrected on luminance, but every contrast gain also amplifies noise; noise is therefore reduced *before* increasing contrast. The steps are: (i) color non-local means, $h = h_{\text{color}} = 20$, patch and search windows 7 and 21; (ii) conversion to Lab and stretching of L on percentiles 1–99; (iii) CLAHE on L with `clipLimit` = 1.5 and 8×8 grid; (iv) Gaussian blur (5×5 kernel, σ = 2) of the a and b channels to reduce chroma noise; (v) fixed chroma gain of 1.5 around neutral (128) to compensate for the saturation lost with the contraction. The gain is not computed from α, which in a real scenario is not known. Denoising reduces noise but also smooths fine details, and too high a chroma gain can give unnatural colors. A bilateral filter on L appears in the code only as a commented-out line and is not used in the final version. All parameters were set empirically and remain the same for every image.

Metrics. PSNR in dB, $10 \cdot \log_{10}(255^2/\text{MSE})$ (higher is better); scikit-image SSIM over the three channels with `data_range=255` (higher is better); mean absolute error MAE (lower is better). All are computed with respect to the original, not the degraded image.

4. Results

Figure 2 compares the original, the degraded image and the three methods on the five images (the Baseline coincides with the degraded image, so it appears only once). Panels are shown at the same scale for each row; full-resolution figures are in the repository. Table 2 reports the metrics.

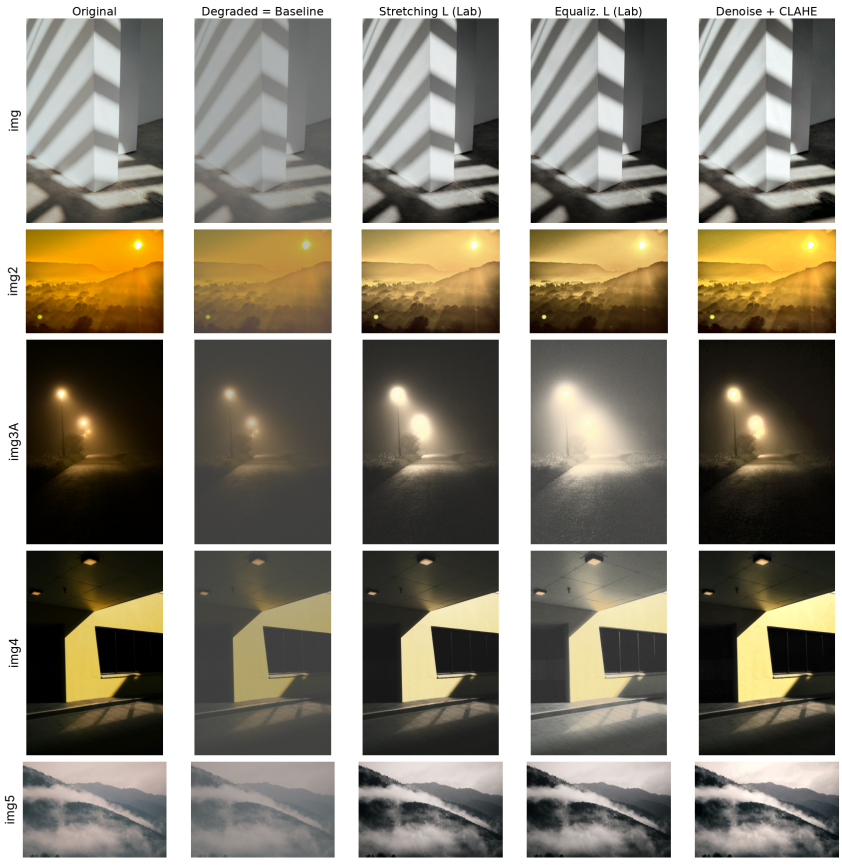


Figure 2 *– Visual comparison on all images: original, degraded (= Baseline), L stretching, L equalization, Denoise + CLAHE.*

***Table 2** – Metrics with respect to the original (PSNR, SSIM, MAE) for each image and method; the mean is the arithmetic mean over the five images. The full CSV is in* `results/risultati_metriche.csv`.

Metric	Method	img	img2	img3A	img4	img5	Mean
PSNR (dB) ↑	Baseline	18.40	14.60	13.06	13.50	20.32	15.98
	Stretching L	19.86	13.07	13.37	17.62	17.20	16.22
	Equalization L	17.36	13.47	6.96	9.58	16.39	12.75
	Denoise + CLAHE	21.23	18.05	21.45	20.07	18.62	19.88
SSIM ↑	Baseline	0.417	0.299	0.144	0.228	0.375	0.293
	Stretching L	0.244	0.206	0.066	0.232	0.119	0.173
	Equalization L	0.221	0.192	0.046	0.122	0.111	0.138
	Denoise + CLAHE	0.734	0.490	0.559	0.478	0.869	0.626
MAE ↓	Baseline	25.93	41.06	54.25	50.93	20.35	38.50
	Stretching L	21.27	41.32	47.53	27.97	29.37	33.49
	Equalization L	29.82	41.38	101.09	73.04	32.49	55.56
	Denoise + CLAHE	19.77	24.47	16.98	20.20	26.33	21.55

5. Discussion

5.1 Qualitative analysis

Case 1 – img3A (night path with fog); preferred method: Denoise + CLAHE. This is the image that most challenges the global methods. Equalization shifts the background, almost black in the original, toward a light gray: the halo of the street lamps invades half the scene, evident grain appears in the sky and the result is no longer a night scene (PSNR 6.96 dB, MAE 101.1, the worst in the whole table). Stretching lifts the background and makes the asphalt more readable, but inflates the lamps into saturated halos. Denoise + CLAHE preserves the dark atmosphere, makes the lamps more defined and does not amplify noise (PSNR 21.45 dB, SSIM 0.559, MAE 16.98: best values). The price is that the vegetation at the sides remains almost black and the grain of the asphalt is partly smoothed by the denoising.

Case 2 – img4 (yellow wall with sharp shadows); preferred method: Denoise + CLAHE. Stretching improves the metrics relative to the baseline (PSNR from 13.50 to 17.62 dB) but the wall becomes a pale, washed-out yellow: L increases while a and b stay unchanged, so the relative saturation drops. Equalization reveals the ceiling panels and the paving joints, but flattens the contrast between light and shadow, which is the character of the scene, and shifts toward a gray-green (PSNR 9.58 dB). Denoise + CLAHE maintains sharp shadow edges and a yellow close to the original thanks to the chroma gain (PSNR 20.07 dB, SSIM 0.478). The gain is however fixed: on the brightest wall the yellow tends to come out slightly lighter than the reference.

Other images. On *img* all methods recover the shadows of the wall; however, stretching and equalization leave visible grain on smooth surfaces, while Denoise + CLAHE keeps them clean (SSIM 0.734). In the same flat area, however, the grain drops to 1.8 with Denoise + CLAHE, below the 2.8 of the original: denoising also smooths part of the real texture. On *img2* stretching lightens the orange toward a pale yellow and equalization pushes the sky toward olive tones; Denoise + CLAHE better preserves the warm hue, but the sun and the reflection remain overexposed and the sky shows some smoothing blotches.

5.2 When the number does not match the appearance

img5 (mountain fog). This is the most instructive case: the baseline obtains PSNR (20.32 dB) and MAE (20.35) better than all methods, yet it is visually the least readable. The original is already low-contrast by nature, so a contrast increase, useful for visibility, moves the result away from the reference and is penalized. Stretching and equalization produce an almost black-and-white image, with the pinkish tint lost and PSNR lower than the baseline; Denoise + CLAHE makes the fog swirls more distinct while preserving the tonality, but brightens the sky.

Disagreement between metrics. Again on img5, Denoise + CLAHE has the best SSIM overall (0.869) but PSNR and MAE worse than the baseline. A measurement on luminance, made on a version reduced to a quarter of the image, shows that the three methods bring the standard deviation of L from about 43 (original) to 70–73, versus 22 for the degraded image: they exceed the contrast of the original, and

Denoise + CLAHE brightens the sky considerably (mean luminance of the upper band from 196 to 232). This deviation weighs on the squared and absolute error, while SSIM, sensitive to local structure, is favored by the denoising. A second case concerns stretching on *img*: PSNR 19.86 dB versus 18.40 for the baseline and better MAE, but SSIM 0.244 versus 0.417. Stretching also amplifies noise: the standard deviation of the high frequency in a flat area of the wall goes from 6.7 (degraded) to 18.6, about three times, versus 1.8 for Denoise + CLAHE. The mean error drops because the contrast becomes more similar to the original, while SSIM penalizes the grain. No single metric is enough to choose the method.

5.3 Limitations

The degradation is synthetic and simple (linear contraction plus Gaussian noise): it does not reproduce real non-uniform illumination or shadows. The parameters are fixed and may be suboptimal for individual images. The metrics measure closeness to the original, and a low-contrast original is itself an imperfect reference: a successful enhancement may move away from it. The comparison is not completely fair, because only method 4 includes denoising: its advantage in SSIM and PSNR is not attributable to CLAHE alone. An ablation (NLM only, CLAHE only, stretching after denoising) would separate the contributions. Finally, no subjective evaluation with observers was performed.

6. Conclusions

The chosen method is Denoise + CLAHE. Across the five images it obtains the best PSNR in four cases and the best MAE in four, the best SSIM in all five (means: PSNR 19.88 dB versus 15.98 for the baseline; SSIM 0.626 versus 0.293; MAE 21.55 versus 38.50), and visually it better preserves the character of the scene and the colors, especially on img3A and img4. The only numerical exception, img5, is explained by the low contrast of the original. Stretching and equalization, applied to luminance only, make details more visible but amplify noise and, in the case of equalization, strongly alter the overall brightness (mean PSNR 12.75 dB, below the baseline). The choice is therefore based on visual and numerical evidence together, not on a single metric, with the limitation that the chroma gain and the denoising strength are fixed and not adapted to the individual scene.

Reproducibility

Python 3.11; install the dependencies with `pip install -r requirements.txt`; run the whole pipeline with `python main.py`, which reads the images from `image/` and writes degraded images, results, figures and CSV in `results/`. The seed (42) and the degradation parameters are constants at the top of `main.py`. File structure and details are in the repository README.